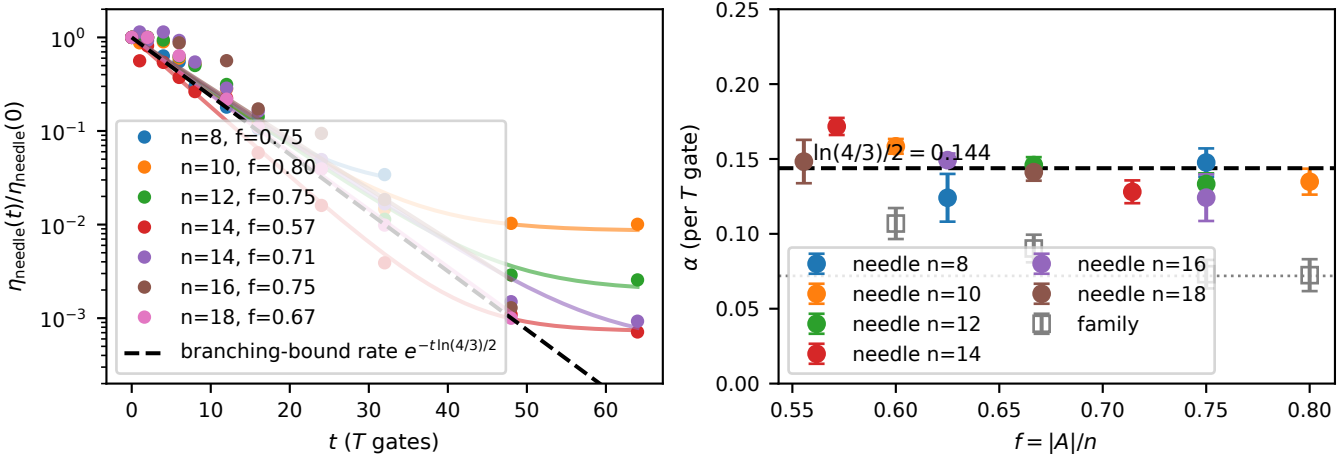

(a) collapse vs branching-bound rate, $n = 8$–18
$\eta_{\text{needle}}(t)/\eta_{\text{needle}}(0)$
t (T gates)
n=8, f=0.75
n=10, f=0.80
n=12, f=0.75
n=14, f=0.57
n=14, f=0.71
n=16, f=0.75
n=18, f=0.67
branching-bound rate $e^{-t\ln(4/3)/2}$
(b) fitted α: needle vs commuting family
α (per T gate)
$f = |A|/n$
$\ln(4/3)/2 = 0.144$
needle n=8
needle n=10
needle n=12
needle n=14
needle n=16
needle n=18
family